Legendre Transform

for thermodynamics and statistical physics

From natural variables to useful potentials

Prepared in English with Awesome Marp

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Why Legendre transforms appear in thermodynamics

Why do we need a new potential?

- A thermodynamic function is most useful when its natural variables are the quantities we can control in the lab.
- But the internal energy is written as $U = U(S, V, N)$, while experiments often fix T, P, or μ instead.
- The Legendre transform exchanges an extensive variable for its intensive conjugate.
- This gives new potentials with simpler differentials and direct physical criteria for equilibrium.

Main idea

> Replace a slope-controlled description by an intercept-like description.

A geometric preview

- For a convex function $f(x)$, define its slope

$$p = \frac{df}{dx}.$$

- The tangent line at x is

$$y = px - \big(px - f(x)\big).$$

- The quantity

$$g(p) = px - f(x)$$

depends on the slope rather than on the original coordinate.

Interpretation

- x is the original variable
- p is the conjugate variable
- $g(p)$ stores the same information in a new coordinate system

For convex f, the map between x and p is one-to-one, so the transform is reversible.

The mathematical construction

Legendre transform in one variable

Given a differentiable convex function $f(x)$,

$$p = \frac{df}{dx}, \qquad g(p) = px - f(x).$$

Then

$$\frac{dg}{dp} = x.$$

- The new independent variable is p instead of x.
- The transform subtracts the term needed to remove dx from the differential.
- In thermodynamics, this is exactly how we move from U to F, H, G, or Ω.

Differential viewpoint: the thermodynamic version

Start from the fundamental relation for a simple system:

$$dU = T\,dS - P\,dV + \mu\,dN.$$

So the conjugate pairs are

- $T \leftrightarrow S$
- $-P \leftrightarrow V$
- $\mu \leftrightarrow N$

If we want to replace S by T, define

$$F = U - TS.$$

Then

$$dF = -S\,dT - P\,dV + \mu\,dN.$$

Now F has natural variables (T, V, N).

Conjugate pairs and sign conventions

General form	Thermodynamic meaning
$dU = T\,dS - P\,dV + \mu\,dN$	fundamental differential
$T = (\partial U/\partial S)_{V,N}$	conjugate to entropy
$-P = (\partial U/\partial V)_{S,N}$	conjugate to volume
$\mu = (\partial U/\partial N)_{S,V}$	conjugate to particle number

The minus sign for pressure comes from mechanical work $\delta W = P\,dV$, so $dU = \delta Q - \delta W$ gives $-P\,dV$.

Thermodynamic potentials from Legendre transforms

The standard family of potentials

Potential	Definition	Natural variables
Internal energy U	U	(S, V, N)
Helmholtz free energy F	$U - TS$	(T, V, N)
Enthalpy H	$U + PV$	(S, P, N)
Gibbs free energy G	$U - TS + PV$	(T, P, N)
Grand potential Ω	$U - TS - \mu N$	(T, V, μ)

Each new potential is chosen to match a common experimental control set.

Reading conjugate pairs from differentials

$$dF = -S\,dT - P\,dV + \mu\,dN,$$
$$dH = T\,dS + V\,dP + \mu\,dN,$$
$$dG = -S\,dT + V\,dP + \mu\,dN.$$

This immediately gives

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}, \qquad P = -\left(\frac{\partial F}{\partial V}\right)_{T,N},$$

and similarly for H and G.

> Useful rule: the coefficient of a differential is the conjugate variable.

A compact map of Legendre moves

Starting from $U(S, V, N)$

- transform in $S \to F(T, V, N) = U - TS$
- transform in $V \to H(S, P, N) = U + PV$
- transform in S and $V \to G(T, P, N) = U - TS + PV$
- transform in S and $N \to \Omega(T, V, \mu) = U - TS - \mu N$

Legendre transforms can be done one variable at a time or several at once.

Chemical potential and multi-component systems

What is chemical potential?

For a one-component system,

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} = \left(\frac{\partial G}{\partial N}\right)_{T,P}.$$

- It is the energy cost of adding one particle while holding the specified variables fixed.
- It controls particle exchange, just as temperature controls entropy exchange and pressure controls volume exchange.
- Its conjugate extensive variable is the particle number N.

Multi-component systems

For several species,

$$dU = T\,dS - P\,dV + \sum_i \mu_i\,dN_i.$$

and therefore

$$dG = -S\,dT + V\,dP + \sum_i \mu_i\,dN_i.$$

- Each species has its own chemical potential μ_i.
- Diffusion and chemical reaction are driven by differences in chemical potential.
- In equilibrium at fixed T and P, matter redistributes until the appropriate μ_i balances are satisfied.

Why the grand canonical ensemble uses μ

- In statistical mechanics, a system in contact with a heat and particle reservoir has fixed T, V, and μ.
- The matching thermodynamic potential is the grand potential

$$\Omega = U - TS - \mu N = F - \mu N.$$

- Its differential is

$$d\Omega = -S\,dT - P\,dV - N\,d\mu.$$

- This is the ensemble-level version of the same Legendre-transform logic.

Another example: electric work

Electric potential and charge

Suppose a system can also exchange charge Q.

An electric-work contribution may appear as

$$dU = T\,dS - P\,dV + \mu\,dN + \Phi\,dQ,$$

where Φ is electric potential.

Then Φ and Q are another conjugate pair.

If Φ is externally controlled, define

$$\widetilde{U} = U - \Phi Q.$$

Its differential contains $-Q\,d\Phi$, so the natural variable becomes Φ instead of Q.

Takeaways

The main message

Remember these points

- A Legendre transform replaces a variable by its conjugate slope.
- In thermodynamics, it converts $U(S, V, N)$ into potentials adapted to fixed T, P, or μ.
- Known conjugate pairs include (T, S), (P, V) with the thermodynamic sign convention, and (μ, N).
- Chemical potential is the variable conjugate to particle number and becomes central for multi-component systems and the grand canonical ensemble.
- The same logic also applies beyond mechanical work, for example to electric potential and charge.

Questions?

Legendre transforms connect mathematics, thermodynamics, and statistical physics.